

JSC370 Final Project

Justin Won

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Introduction

Crime rates vary significantly across countries and it is important to understand the factors that influence crime for economics and public policy. One key variable that is often linked to crime is income level; specifically, whether higher income levels reduce crime by increasing economic opportunities or conversely, whether income inequality increases criminal activity. Additionally, education levels have been studied as another determinant of crime since higher education is associated with better employment prospects and lower engagement in criminal activity.

This analysis investigates the relationship between average income (GDP per capita in US dollars), education levels (literacy rate), and crime rates (intentional homicides per 100,000 people) across multiple countries. I will be using data retrieved from the World Bank API and I will analyze whether countries with higher income and education levels tend to have lower crime rates.

In addition to descriptive visualizations and correlation analysis, I use statistical tests such as the Shapiro-Wilk test to assess normality and apply log-transformations where appropriate. I also conduct multiple linear regression and weighted regression to better model the relationship between predictors and crime rates while addressing issues such as skewness and heteroscedasticity. Furthermore, I perform model comparison using ANOVA and AIC criteria to evaluate whether the inclusion of literacy rate improves model performance beyond income alone. These methods allow for a more rigorous interpretation of how economic and educational factors relate to crime across different national contexts.

I would like to contribute to the broader discussion on how economic and educational policies can be leveraged to reduce crime by conducting this analysis. My findings may provide insights for people seeking to implement strategies that promote safer societies through economic development and educational investments.

Methods

Method Part 1: Data Acquisition and Wrangling

The data was retrieved using API queries in R with the `httr` and `jsonlite` packages. I found the data for this analysis from the World Bank API, which provides publicly accessible economic and social indicators for countries worldwide in 2015. These three indicators were used: Income in GDP per capita using `NY.GDP.PCAP.CD` which measures the economic output per person in each region/country, crime rate in Intentional Homicides per 100,000 using `VC.IHR.PSRC.P5` which represents the number of homicides per 100,000 individuals and education Levels in Adult Literacy Rate using `SE.ADT.LITR.ZS` which measures the percentage of literate adults in each country.

First, I got the API URL then fetched a dataset which contains regional aggregates/country code, year 2015, and income (GDP per capita) for each. Then, after fetching the data, I wanted to turn it into a dataframe, so I got the JSON format to rename the variables so that we can understand the variables more clearly. Then, I removed the N/A values for income and regional aggregates/country code to clean out the data. Similarly, I got the API URL then fetched a dataset which contains regional aggregates/country code, year 2015, and

crime rate (Intentional Homicides per 100k people) for each. Then, after fetching the data, I wanted to turn it into a dataframe, so I got the JSON format to rename the variables so that we can understand the variables more clearly. Then, I removed the N/A values for crime rate and regional aggregates/country code to clean out the data. Similarly, I got the API URL then fetched a dataset which contains regional aggregates/country code, year 2015, and literacy rate for each. Then, after fetching the data, I wanted to turn it into a dataframe, so I got the JSON format to rename the variables so that we can understand the variables more clearly. Then, I removed the N/A values for the regional aggregates/country code to clean out the data, but not the literacy rate since I wanted a smooth merging process. Finally, I merged these three datasets with country_code and year. This would be equivalent to a left join.

Method Part 2: Exploratory Analysis and Preliminary Results

I used kable for the summary statistics table which includes mean, median and sd of each variables. Also, tables listing high and low values of the key variables were created. North American region has the highest income on average and Afghanistan has the lowest income on average in 2015. Caribbean small states has the highest crime rate on average and Europe & Central has the lowest crime rate on average. Central Europe and Baltic region has the highest literacy rate on average and Afghanistan has the lowest literacy rate on average in 2015.

Distribution Histograms and Log Transformation

Then, I explored the histogram to check the frequency distribution of data values. Looking at the income histogram, we can see that the data is highly right-skewed/positively skewed. A small number of countries have very high GDP per capita, pulling the distribution to the right. The majority of countries have low or moderate incomes which creates a long tail towards higher values. This means that using raw income values in this analysis might not be appropriate due to the outliers. The crime rate distribution is also right-skewed. Most countries have low crime rates, but a few have exceptionally high crime rates. This means that a small number of countries with high crime rate is dominating the dataset which could violate normality. As stated, transformations like $\log(\text{crime_rate})$ can help normalize the distribution. However, the literacy rate appears less skewed with values clustered towards the higher end above 70-80%. Unlike income and crime rate, literacy rates seems to be normally or slightly left-skewed. Most countries have high literacy rates, with fewer having low literacy levels. Both the density plots for income and crime rate confirmed that both datasets are right skewed, so we will continue this analysis using log transformed income and crime rate. As expected, after log transforming income and crime rate, it seems more normally distributed.

Scatterplots

A scatterplot is used to visualize the relationship between log-transformed income and log-transformed crime rate. A negative trend and correlation is observed which means that as income increases, crime rates tend to decrease. As mentioned previously, we used the log transformed variables since both the data for income and crime rates were highly right skewed. This reduces the effect of outliers, allows for a clearer trend in the relationship and makes interpretation more meaningful in percentage changes. As seen on the plot, countries with higher income generally have lower crime rates. However, the spread of points shows that crime rates vary significantly even among countries with similar income levels which could mean that other factors such as education, governance and inequality might also influence crime rate. In general, countries with lower $\log(\text{income})$ on the left side shows higher and more dispersed crime rates. Some have crime rates above $\log(3.0)$ which translates to over 20 homicides per 100,000 people. Countries with higher $\log(\text{income})$ on the right side tend to cluster at lower crime rates supporting the hypothesis that wealthier nations experience lower crime. However, a few outliers exist, showing some high-income nations with unexpectedly high crime rates.

Another scatterplot is used to show the relationship between literacy rate and the $\log(\text{crime_rate})$. The regression line shows a slight negative correlation which means that countries with higher literacy rates tend

to have lower crime rates. However, the correlation seems to be bit weaker than the previous plot with a noticeable variation in crime rates even among countries with similar literacy rates. Countries with lower literacy rate have higher and more dispersed crime rates with some above $\log(3.0)$ which corresponds to more than 20 homicides per 100,000 people. Countries with higher literacy rate seems to have generally lower and more clustered crime rates with few outliers. However, there are some exceptions where high literacy countries still experience relatively high crime rates which means that other factors such as income inequality, governance, or social instability might also play a role. In summary, a higher literacy rate might contribute to lower crime rates due to more economic opportunities and reduced poverty driven crimes, but the spread of data points shows that literacy alone does not fully explain crime rates.

Classifications and Boxplots by Income & Literacy Category

For this part, we will be using the classifications from 2015 written on the worldbank website to divide the categorical variables from: <https://blogs.worldbank.org/en/opendata/new-country-classifications#:~:text=As%20of%201%20July%202015,GNI%20per%20capita%20of%20%2412%2C736>.

The largest category consists of Upper-Middle Income countries which make up the highest number in this dataset. Also, a significant portion of the analyzed countries are categorized as Lower-Middle Income countries. We can also see that there are only few high and low income countries. Looking at the income boxplot, High-Income Countries have the lowest median crime rate and a smaller spread which means less variability in crime levels. We can see that Low-Income Countries show higher median crime rates showing a possible link between economic hardship and crime. Lower-Middle and Upper-Middle Income Groups have higher crime rates and larger variability meaning that crime levels fluctuate more within these groups. The longer whiskers and spread-out data in some groups means that some countries have exceptionally high or low crime rates relative to their income level. We may need further analysis to explore whether factors like income inequality or education levels contribute to crime patterns. Low-Literacy Countries tend to have higher crime rates with a higher median and bigger spread than high-literacy countries. High-Literacy Countries have a lower median crime rate and less variation though a few outliers exist. The “NA” category represents countries with missing literacy data which shows a wider distribution of crime rates. We did not clean this category since we are working with a small dataset.

A density plot is used to visualize crime rate by group. In the green region meaning Low-Income Countries, the crime rate distribution is highly concentrated around $\log(\text{Crime Rate}) \approx 2.0$. This shows that most low-income countries have similar crime rates. In the light blue region meaning Lower-Middle Income, the crime rate distribution is broader with peaks around $\log(\text{Crime Rate}) \approx 1.5$ to 2.0 . There is more variation in crime rates among lower-middle-income countries compared to low-income countries. In the purple region which represents the Upper-Middle Income, it shows a wider spread in crime rates with some countries having very low crime rates ($\log \approx 0.5$) and some with high crime ($\log > 3$). The distribution is more evenly spread compared to Low-Income or Lower-Middle income countries. In the red/pink region which are High-Income countries, the crime rate distribution is more dispersed and generally lower with a peak closer to $\log(\text{Crime Rate}) \approx 0.5$ to 1.0 . This suggests that crime rates fluctuate significantly within this group potentially due to external factors like inequality, governance, or urbanization.

Spearman Correlation Heatmap

The Spearman method is used to analyze the strength and direction of Spearman correlations between $\log(\text{income})$, $\log(\text{crime_rate})$, and literacy_rate . $\log(\text{Income})$ and Literacy Rate ≈ 0.904 which means a strong positive correlation. It means that higher-income countries tend to have higher literacy rates. This makes sense because wealthier nations invest more in education, infrastructure, and public services, leading to improved literacy levels. This relationship is the strongest in the matrix implying that economic growth is strongly linked to education. $\log(\text{Income})$ and Crime Rate ≈ 0.243 which means a weak negative correlation. This means that higher-income countries tend to have lower crime rates. However, the correlation is weak meaning that while income might influence crime, other factors such as inequality, governance, or social policies plays a significant role. This aligns with previous boxplots and density plots where Low-Income

countries had consistently higher crime rates, but variation existed in Middle-Income and High-Income countries. Literacy Rate and Crime Rate ≈ -0.312 which means a Weak to moderate negative correlation. This means that countries with higher literacy rates tend to have lower crime rates. While this relationship is stronger than income-crime, it is still not very strong. As explained, Other factors may be involved in this variation.

We use the Spearman Correlation heatmap to visualize Spearman correlations between $\log(\text{income})$, $\log(\text{crime_rate})$, and literacy_rate . Income and Literacy Rate is 0.86 which shows a strong positive correlation. This suggests that higher-income countries tend to have higher literacy rates. This relationship is expected, as wealthier countries typically invest more in education, leading to higher literacy levels. Income and Crime Rate is -0.24 which shows a weak negative correlation. This suggests that higher-income countries tend to have lower crime rates, but the relationship is not strong. This implies that income alone does not fully explain crime rates, and other factors play a role. Literacy Rate and Crime Rate is -0.40 which is a moderate negative correlation. This shows that higher literacy rates tend to have lower crime rates. While literacy appears to have a stronger negative correlation with crime than income, the relationship is still not definitive, meaning other factors might also influence crime rates as explained.

Method Part 2: Statistical analysis

Shapiro-wilk test

The Shapiro-Wilk test checks whether a variable follows a normal distribution. If p-value is greater than 0.05, we fail to reject the null hypothesis, so the data appear normal. If p-value is less than 0.05, we reject the null hypothesis, so the data do not appear normal. By looking at the result, the log-transformed income and log-transformed crime rate both pass the normality test, so they are approximately normal and suitable for models assuming normality such as linear regression. However, literacy rate is not normally distributed ($p < 0.05$). This could impact methods sensitive to normality.

Residual Diagnostics

We use the Diagnostic plots from the basic regression model. Looking at the residual vs fitted, we would want the residuals to be randomly scattered around the horizontal line with no patterns. However, in this plot, there's a mild curve in the residuals, especially on the left which suggests non-linearity or mild model misspecification. Looking at the Q-Q plot, we would want the data points to fall roughly along the 45° reference line which would mean normal residuals. However, we see deviations at the tails, especially a few points on the right. This suggests some departure from normality, but it's not extreme. Looking at the Scale-Location graph, we want the horizontal red line with equal spread (homoscedasticity). There is a slight funnel shape left to right which indicates mild heteroscedasticity (non-constant variance). Looking at the Residuals vs Leverage plot, we would want no highly influential points which means no observations far from center with high Cook's distance. A few observations stand out, but they have moderate leverage. We can see that there are no points that exceed the Cook's distance threshold, so there are no extreme outliers. In summary, the model is mostly acceptable, but there are some small issues with a slight non-linearity, heteroscedasticity and non-normality of residuals.

Weighted Regression Output

To address the issues seen in the residual diagnostics, we are going to perform weighted regression. $1/\text{income}^2$ was chosen as the weight since this adds more weight to lower-income countries and less to countries with very high income. By doing this, variance would be lowered and residuals would be more stable. This helps stabilize variance across fitted values. Looking at the model fit, Adjusted $R^2 = 0.2666$ which means about 26.7% of the variation in $\log(\text{crime rate})$ is explained by $\log(\text{income})$ and literacy rate. Also, we have that $F\text{-statistic} = 7.726$, $p\text{-value} = 0.00166$ which means that the model overall is statistically significant. Looking at the intercept, we see the baseline $\log(\text{crime rate})$ when predictors equal 0. Looking at

the $\log(\text{income})$, the p-value is greater than 0.05 which means that there is no evidence that income predicts crime when controlling for literacy. However, looking at the literacy rate, the p-value is less than 0.05 with -0.0173 as the estimate which means higher literacy rates are associated with lower crime rates. After accounting for heteroscedasticity with weights, literacy rate emerges as the key predictor. The coefficient -0.0173 means that 1 percentage point increase in literacy rate is associated with about 1.7% decrease in crime rate on the log scale. Income becomes statistically insignificant after adjusting for literacy possibly due to collinearity or the dampening effect of weights.

VIF Table

We check for multicollinearity as seen in the weighted regression. VIF is around 5.48 for both predictors indicates moderate multicollinearity. This potentially explain why $\log(\text{income})$ became insignificant when literacy_rate was added.

ANOVA & AIC Model Comparison

To evaluate the impact of adding literacy_rate, we look at the ANOVA model comparison and AIC comparison. In the ANOVA model comparison, adding literacy_rate does not significantly improve the model over using $\log(\text{income})$ alone, based on the F-test. There's insufficient evidence that literacy_rate adds explanatory power in this specific comparison. Looking at the AIC comparison, based on AIC, the simpler model ($\log(\text{income})$ only) is slightly preferred. The inclusion of literacy_rate does not justify the added complexity, at least in terms of predictive performance.

Results

Figure 1: Summary Statistics

Table 1: Summary Statistics (Income & Crime)

mean_income	median_income	sd_income	mean_crime	median_crime
9242.597	5525.005	11993.84	7.406383	5.9

Table 2: Summary Statistics (Literacy)

sd_crime	mean_literacy	median_literacy	sd_literacy
6.314062	78.14366	78.82537	15.46087

Figure 2: Countries with high-low values of the key variables

Table 3: Top 5 Countries by Income

country_code	year	income	crime_rate	literacy_rate
NAC	2015	55702.09	4.6	NA
PST	2015	40778.34	2.3	NA
OED	2015	35784.87	4.5	NA
EMU	2015	34415.62	1.0	NA
EUU	2015	30716.63	1.0	NA

Table 4: Bottom 5 Countries by Income

country_code	year	income	crime_rate	literacy_rate
AFG	2015	565.57	9.98	33.75
HPC	2015	909.49	8.10	60.87
LDC	2015	964.45	6.20	61.78
IDX	2015	1036.87	6.50	64.06
IDA	2015	1346.70	6.90	64.36

Table 5: Top 5 Countries by Crime Rate

country_code	year	income	crime_rate	literacy_rate
CSS	2015	11835.29	31.5	NA
TLA	2015	8617.55	22.7	93.30
LCN	2015	8759.81	22.6	93.43
LAC	2015	8337.15	21.2	93.08
SSF	2015	1628.91	13.7	64.29

Table 6: Bottom 5 Countries by Crime Rate

country_code	year	income	crime_rate	literacy_rate
EMU	2015	34415.62	1.0	NA
EUU	2015	30716.63	1.0	NA
CEB	2015	12568.30	1.3	99.33
EAS	2015	9562.11	1.3	94.93
TEA	2015	6556.50	1.4	95.14

Table 7: Top 5 Countries by Literacy Rate

country_code	year	income	crime_rate	literacy_rate
CEB	2015	12568.30	1.3	99.33
TEC	2015	8001.11	5.7	99.00
ECA	2015	6282.67	6.4	98.42
ECS	2015	22657.31	3.3	98.28
LTE	2015	8228.52	4.8	95.65

Table 8: Bottom 5 Countries by Literacy Rate

country_code	year	income	crime_rate	literacy_rate
AFG	2015	565.57	9.98	33.75
AFW	2015	1845.77	8.10	55.01
PRE	2015	1501.74	8.10	60.48
HPC	2015	909.49	8.10	60.87
LDC	2015	964.45	6.20	61.78

Distribution Histograms

Figure 3: Income

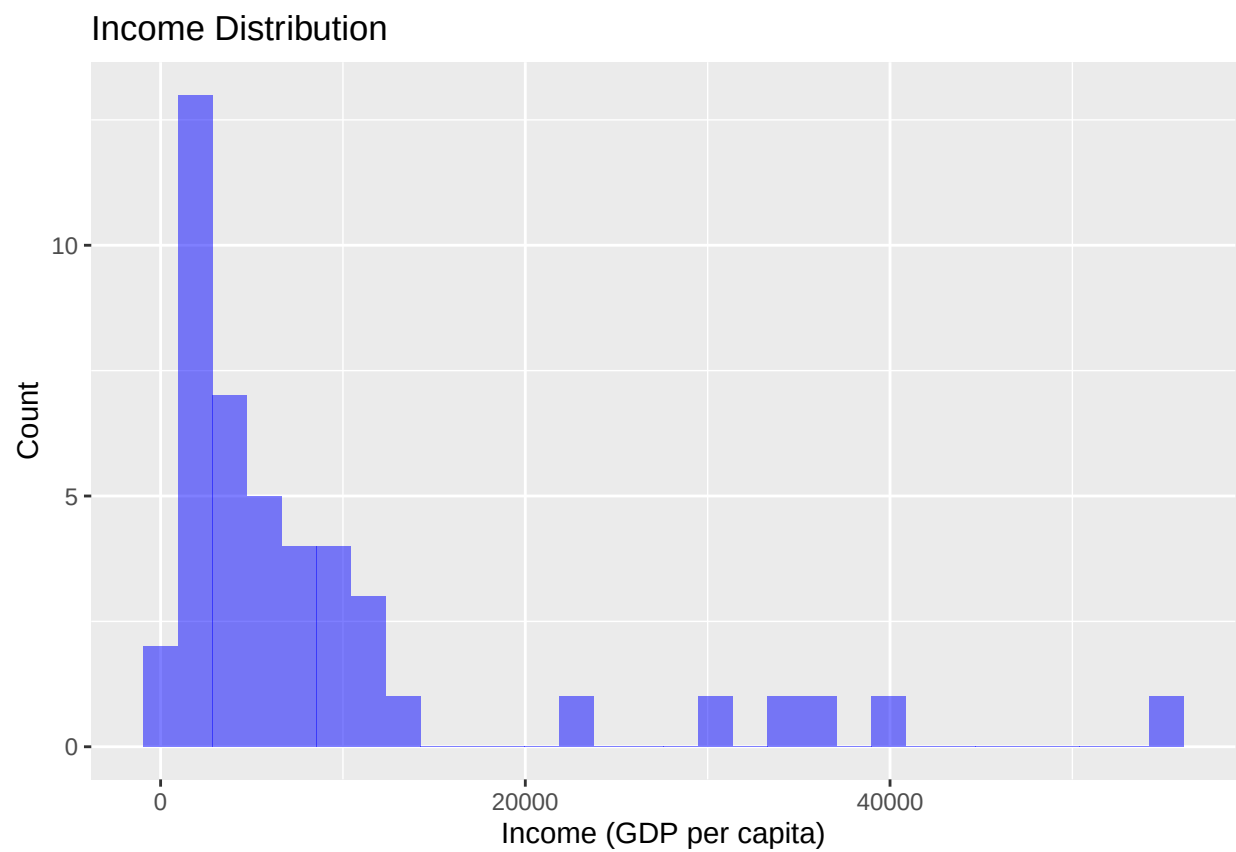


Figure 4: Crime rate

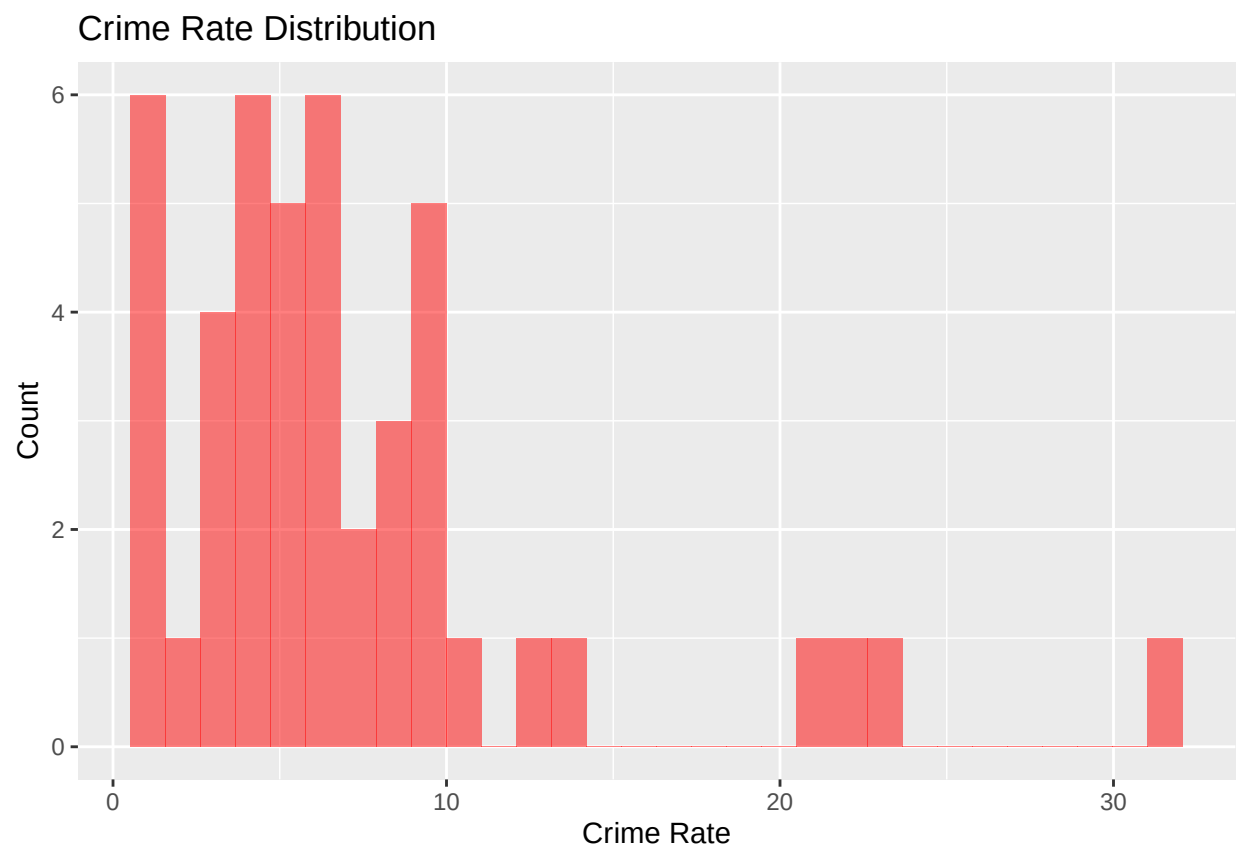


Figure 5: Literacy rate

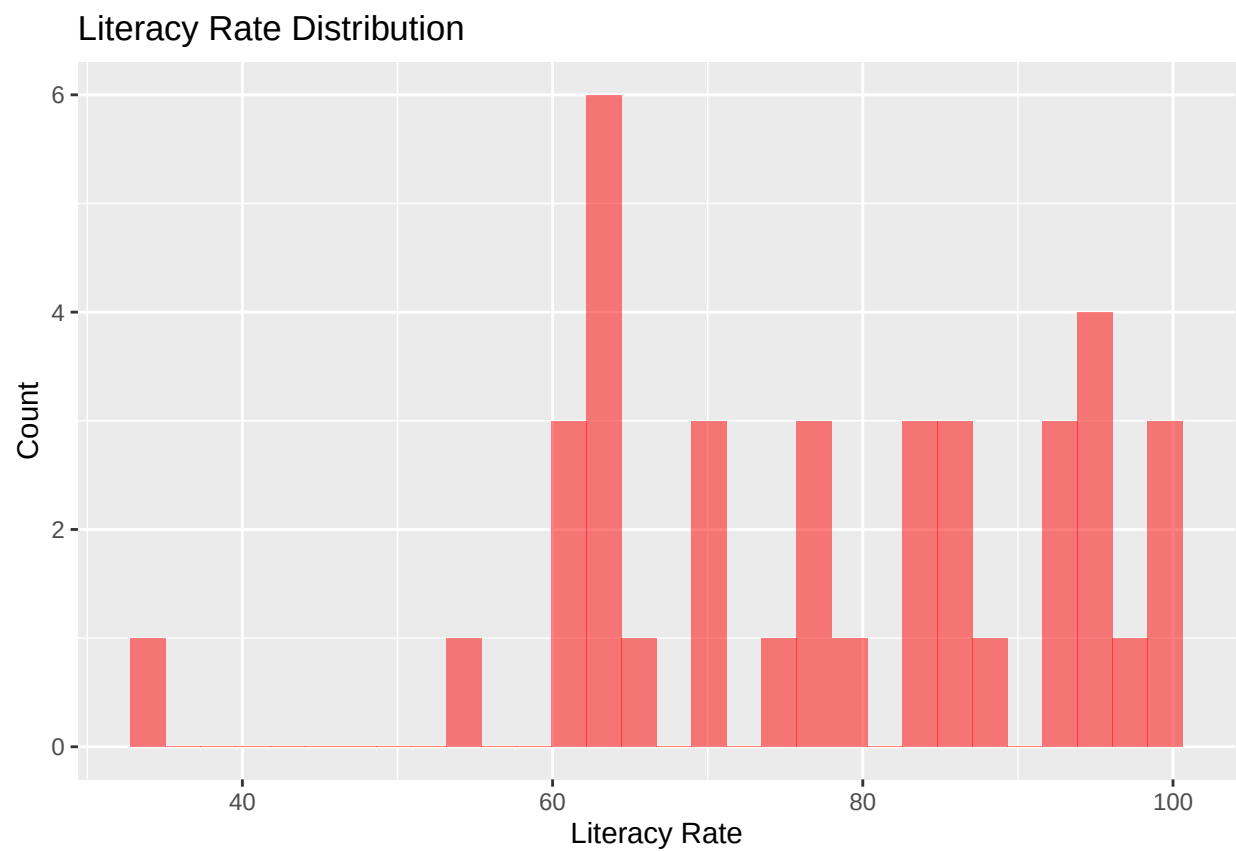


Figure 6: Density plot for Income

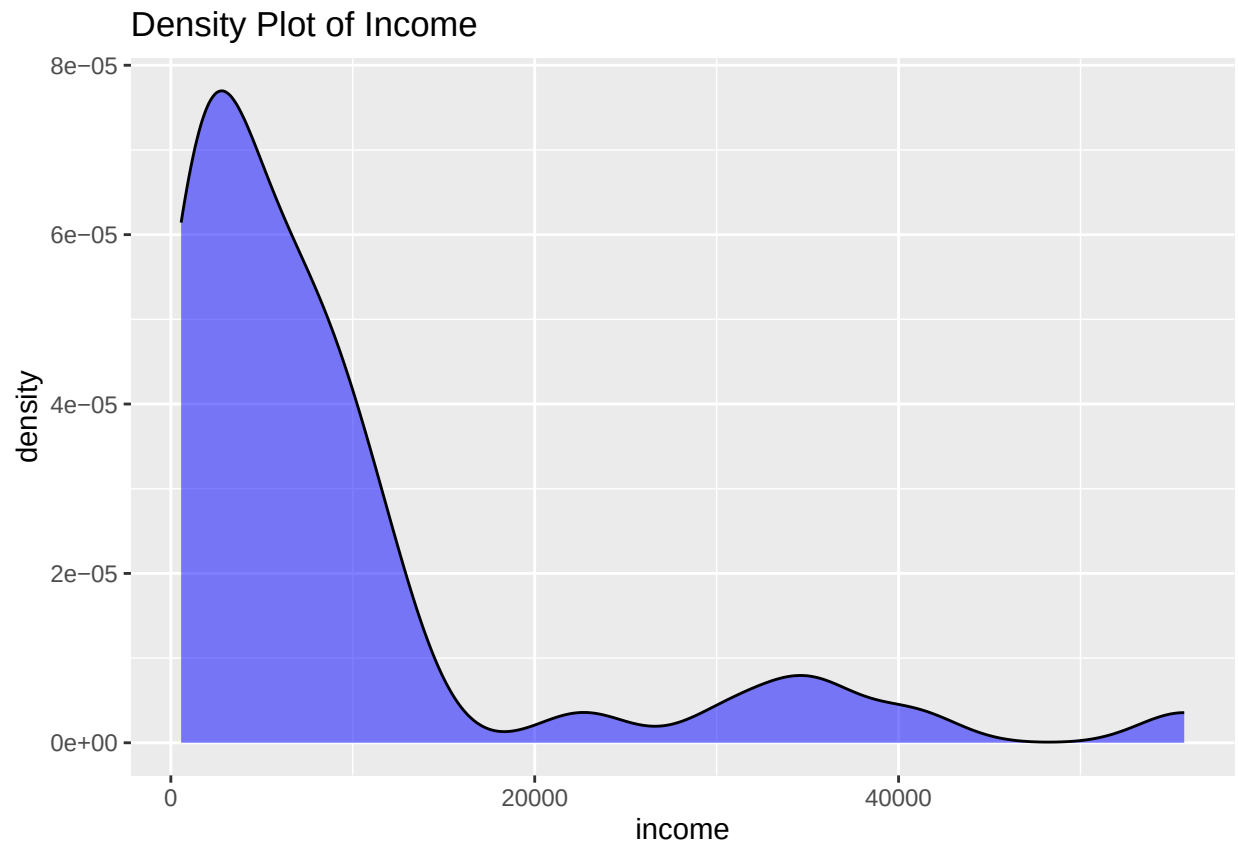


Figure 7: Density plot for Crime rate

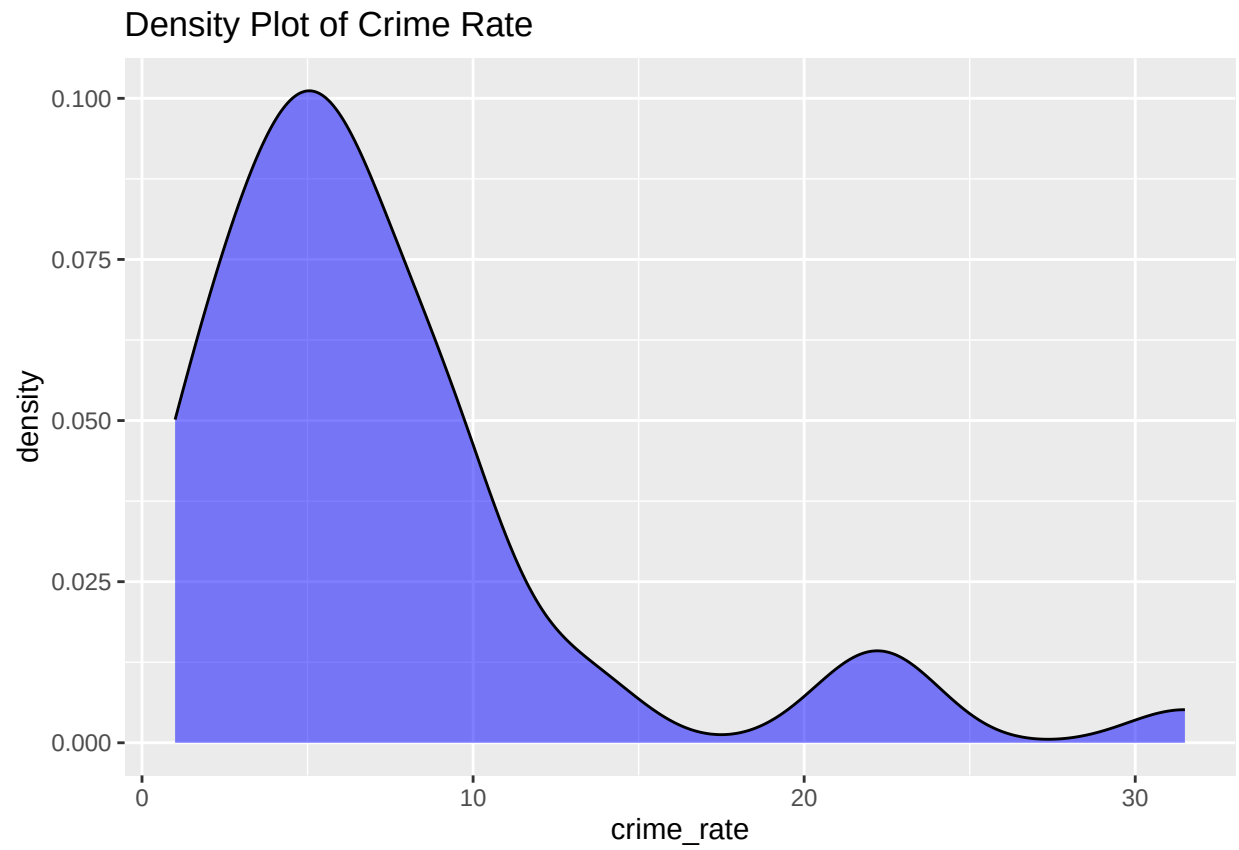


Figure 8: Log-Transformed Income Distribution

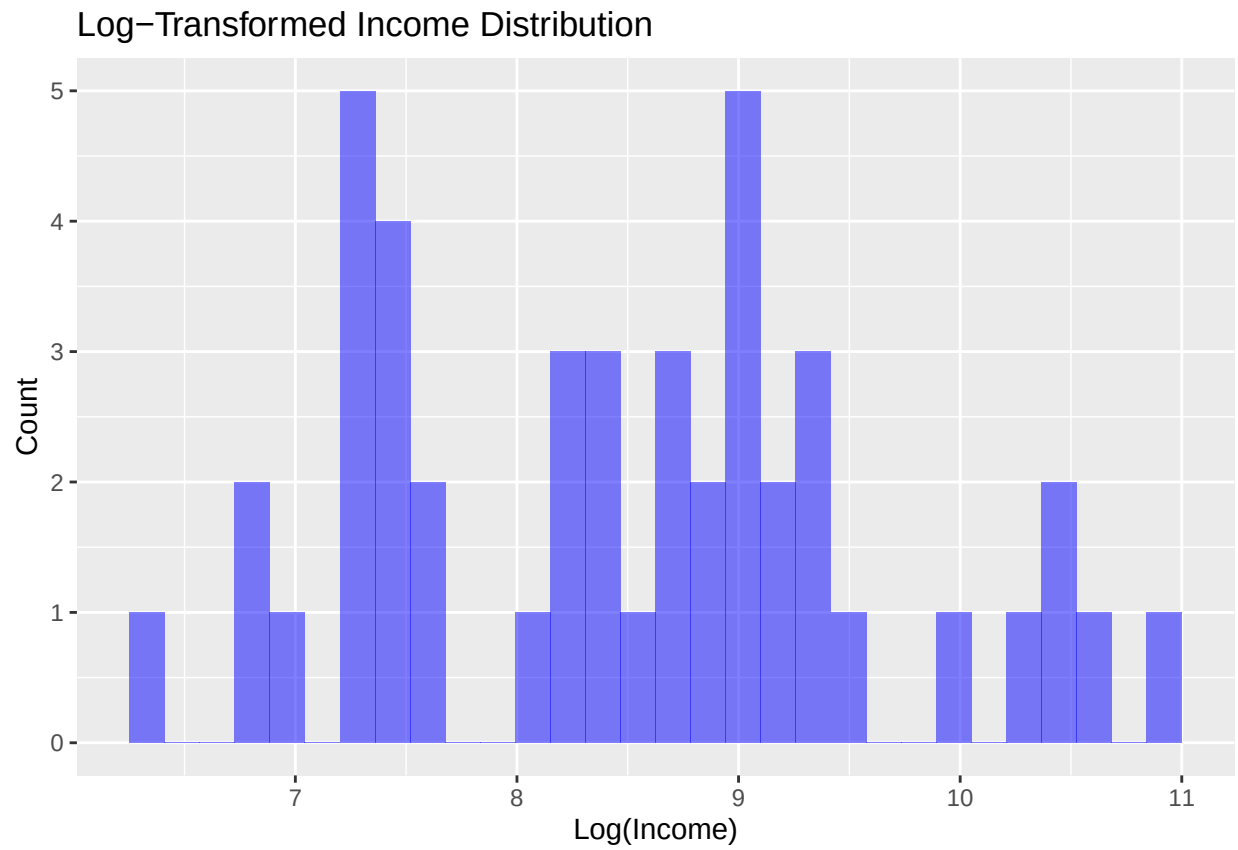


Figure 9: Log-Transformed Crime Rate Distribution

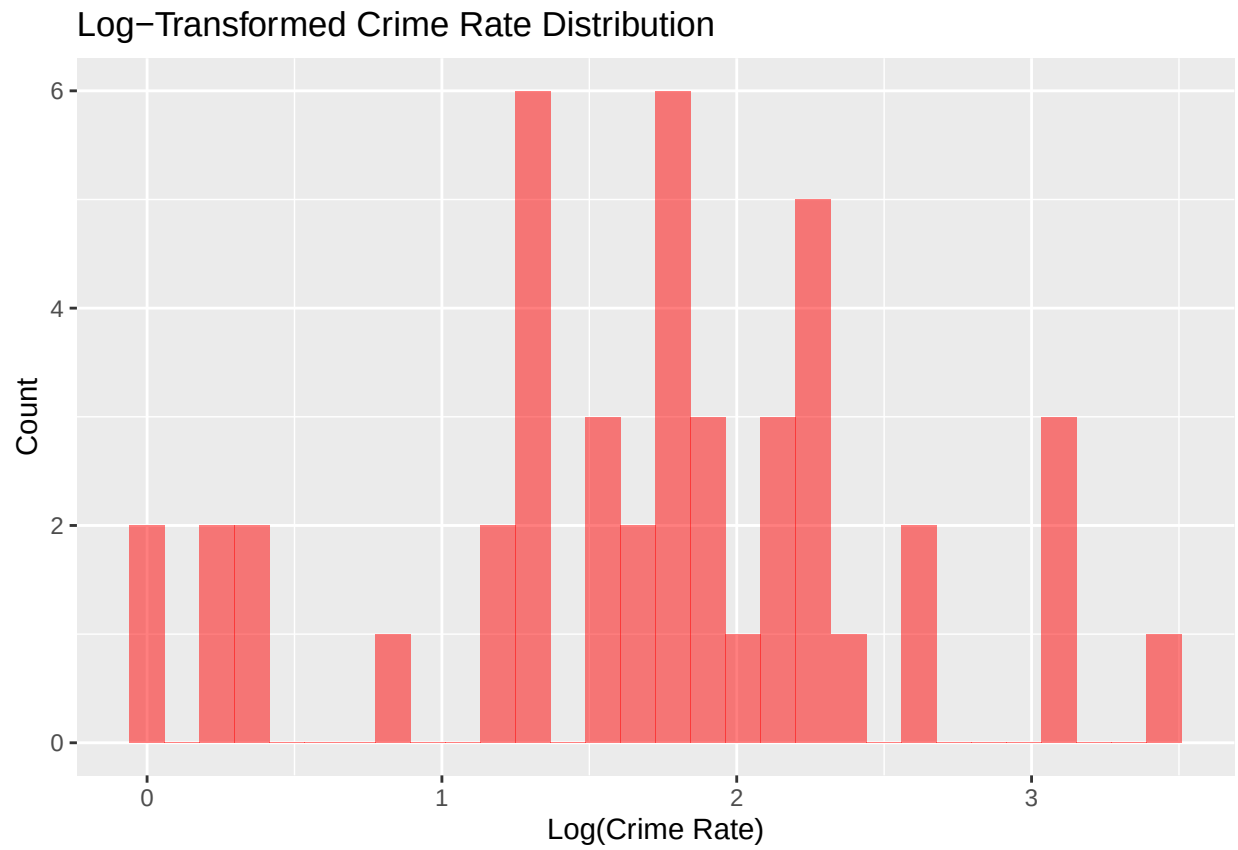
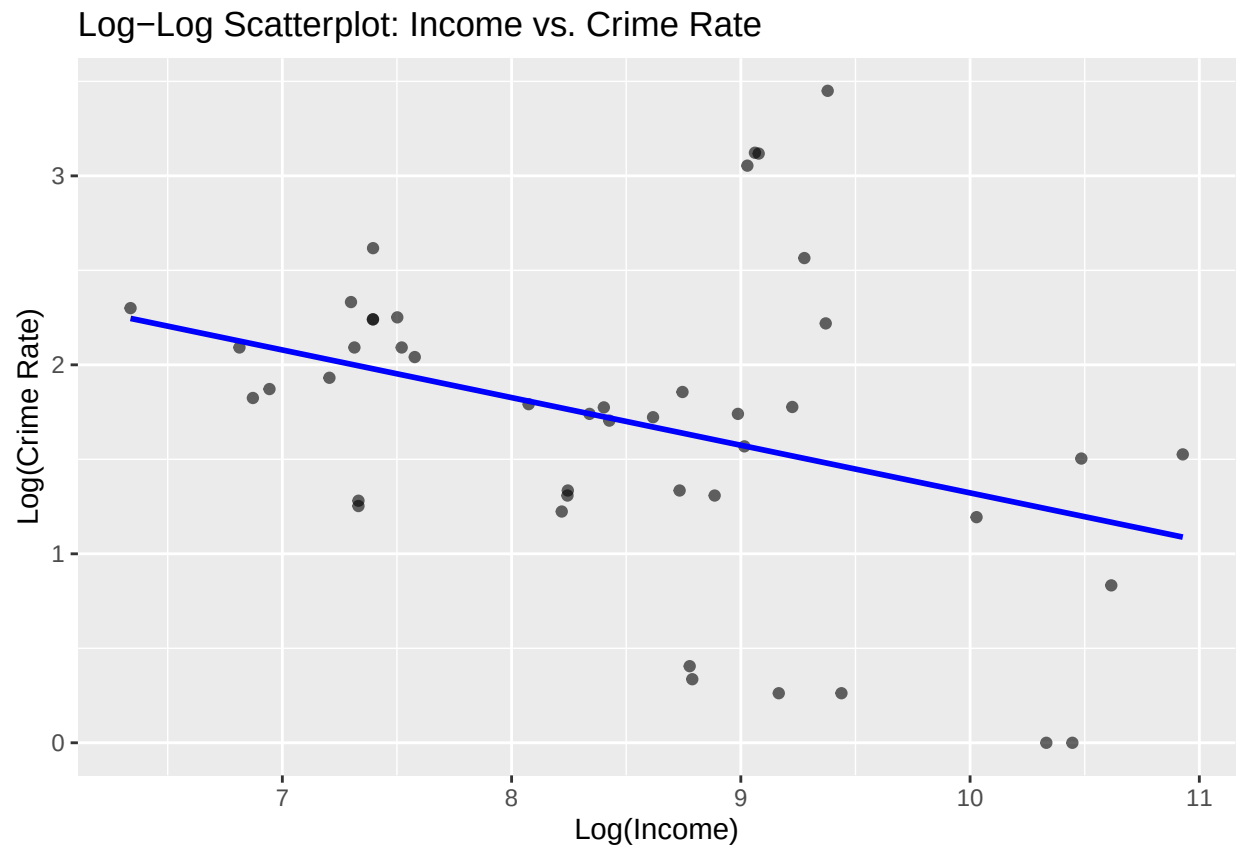
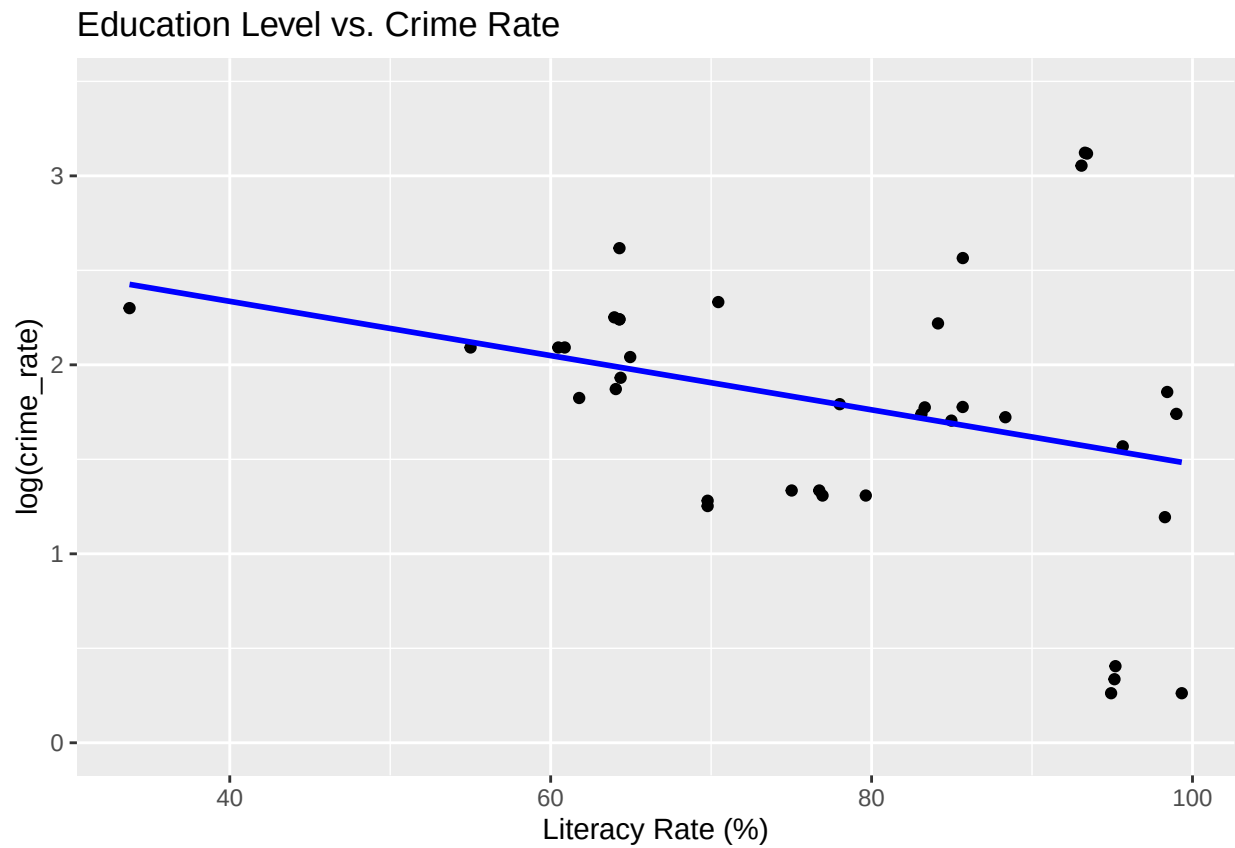


Figure 10: Log-Log Scatterplot: Income vs. Crime Rate



```
## [1] -0.3522739
```

Figure 11: Education Level vs. Crime Rate Scatterplot



```
## [1] -0.3122335
```

Figure 12: Number of Countries in Each Income Group

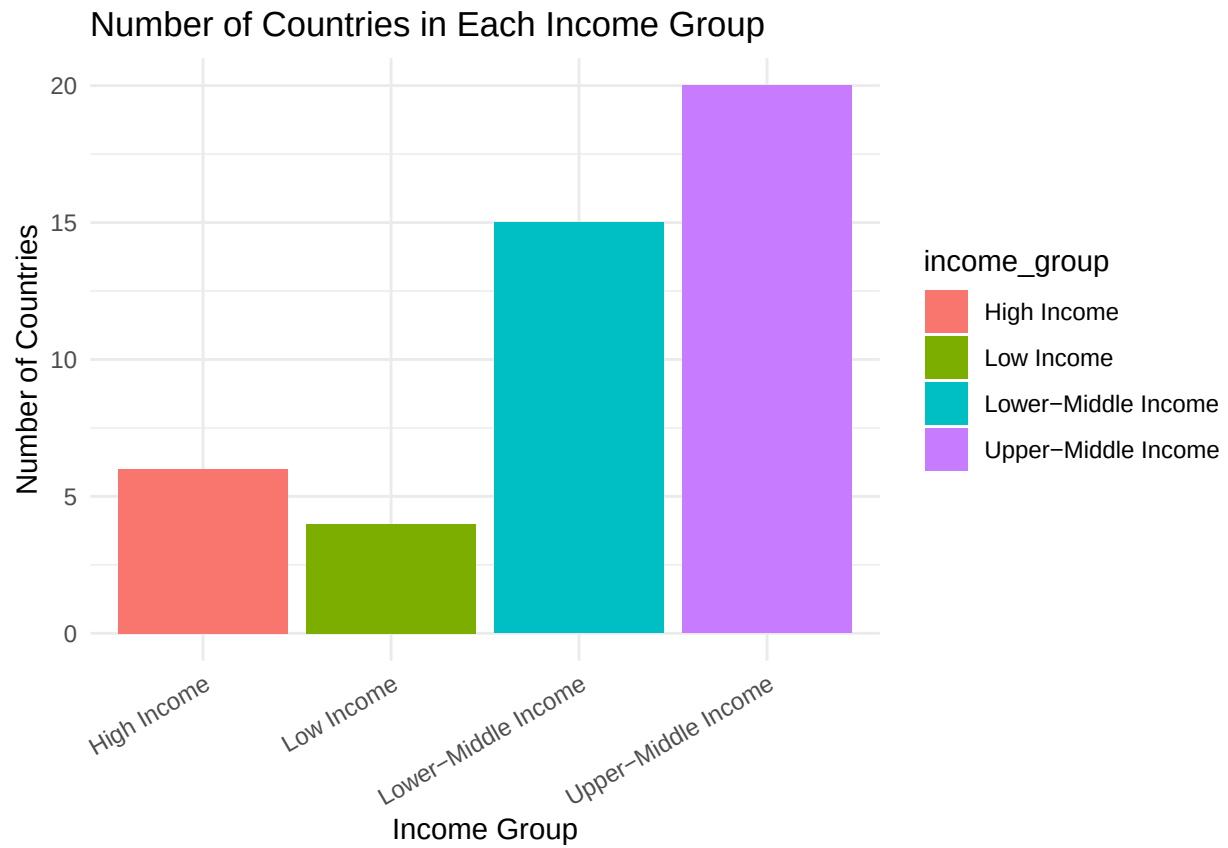


Figure 13: Boxplot: Crime Rate by Income Group

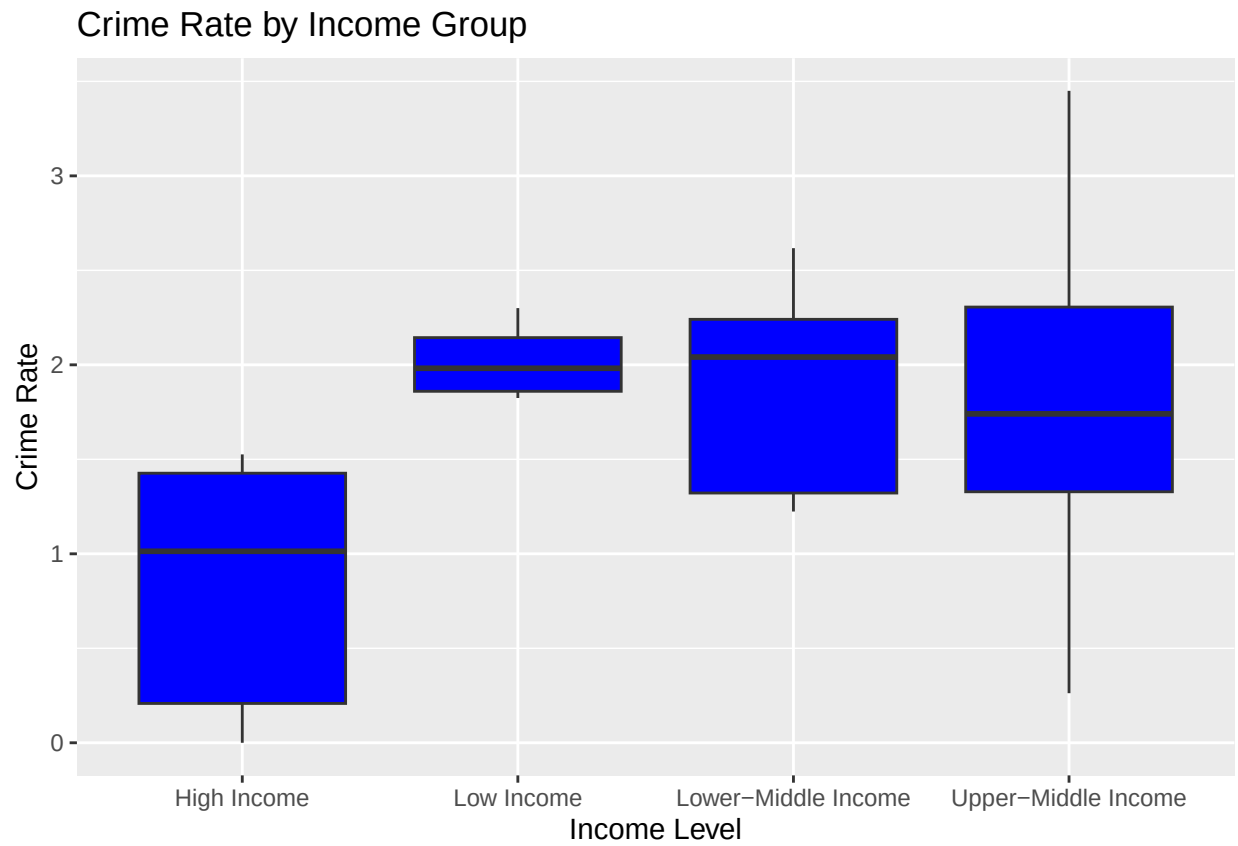


Figure 14: Boxplot: Crime Rate by Literacy Group

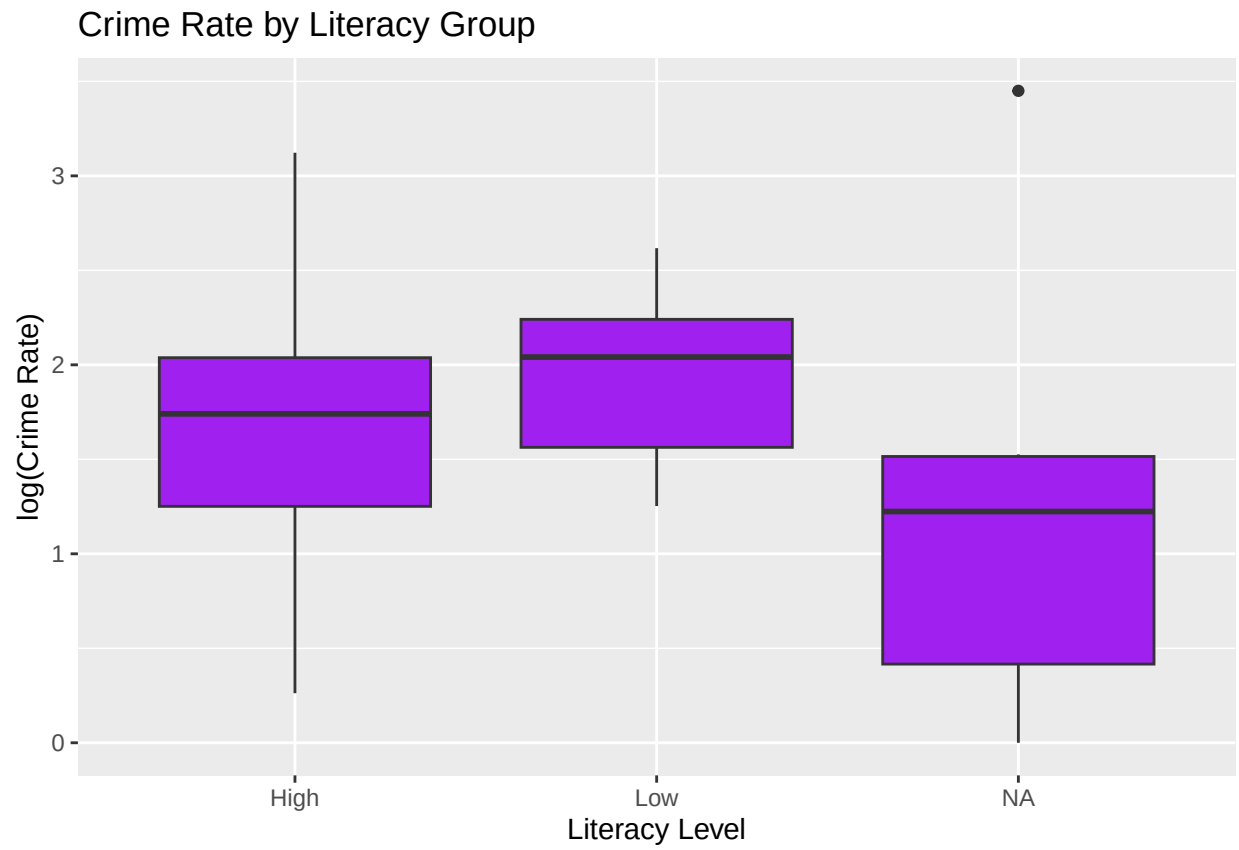


Figure 15: Density Plot: Crime Rate by Income Group

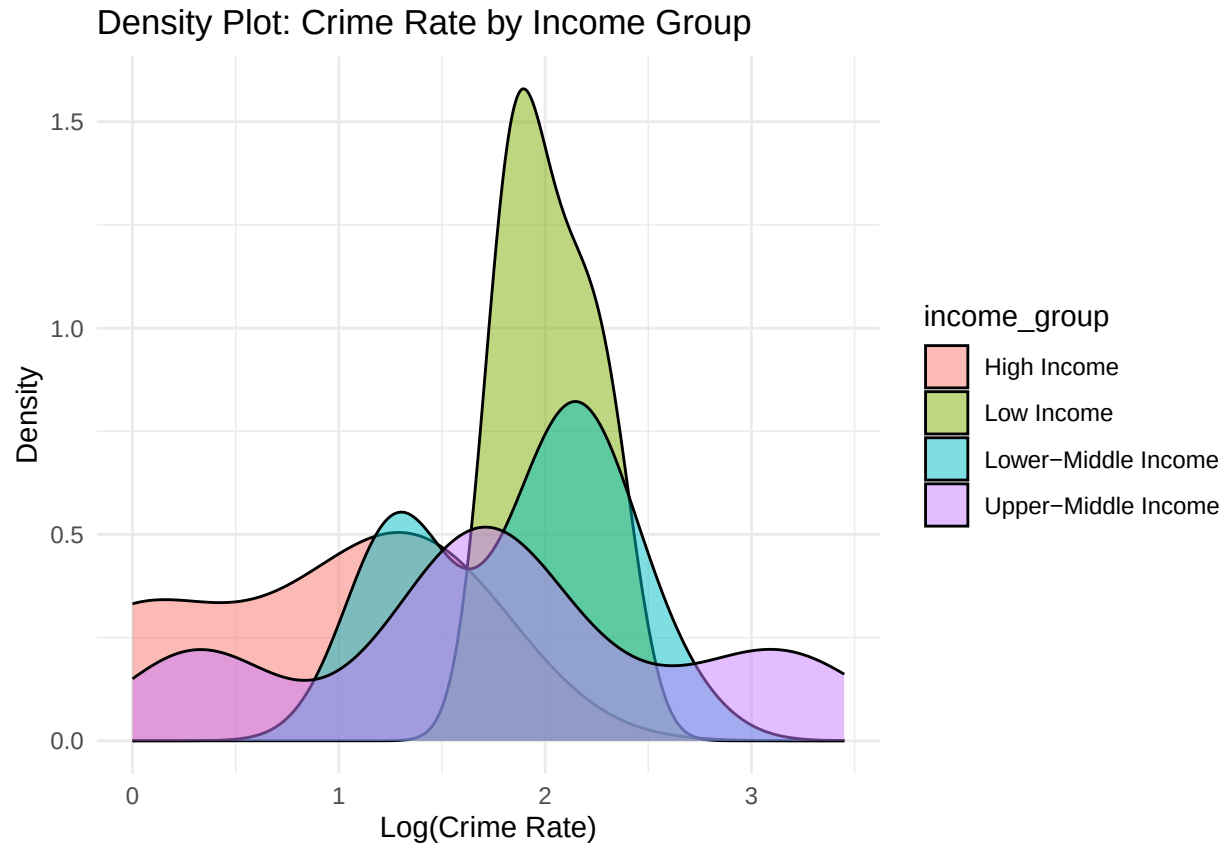


Figure 16: Spearman Correlation Heatmap

```
##          log_income log_crime_rate literacy_rate
## log_income      1.0000000    -0.2434619     0.9042342
## log_crime_rate -0.2434619     1.0000000    -0.3122335
## literacy_rate   0.9042342    -0.3122335     1.0000000
```

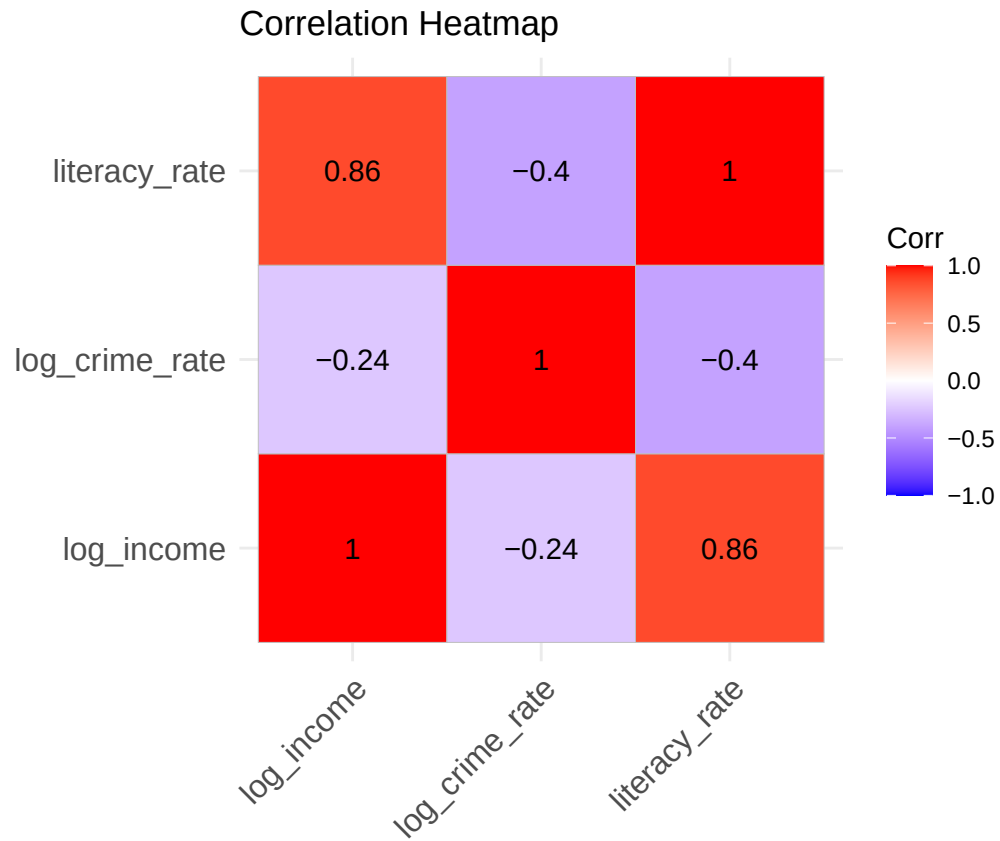


Figure 17: Shapiro-Wilk test

Table 9: Shapiro-Wilk Normality Test Results

Variable	W	p_value
log(income)	0.9651	0.1899
log(crime_rate)	0.9627	0.1543
literacy_rate	0.9317	0.0227

Figure 18: Residual Diagnostics

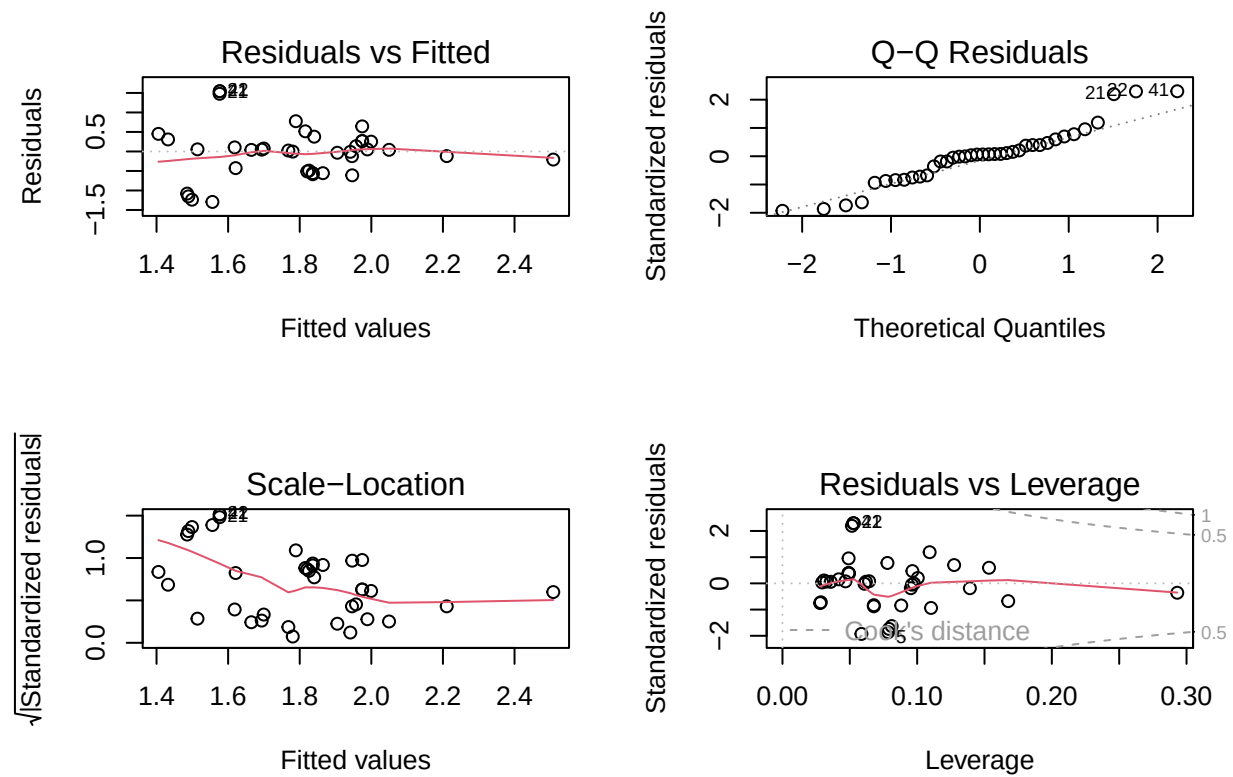


Figure 19: Weighted Regression

Table 10: Weighted Regression Coefficients

Variable	Estimate	Std. Error	t value	p-value
(Intercept)	1.9770	0.9552	2.0698	0.0459
log(income)	0.1493	0.1818	0.8213	0.4170
literacy_rate	-0.0172	0.0068	-2.5469	0.0154

Table 11: Model Fit Statistics

	Adjusted.R.squared	Residual.Std..Error	F.statistic	Model.p.value
value	0.2666	0.000165	7.726	0.001663

Figure 20: VIF Table

```
## log(income) literacy_rate
## 5.483642 5.483642
```

Figure 21: ANOVA & AIC Model Comparison

Table 12: ANOVA Model Comparison: Income vs Income + Literacy

Metric	Value
F-statistic	1.820
p-value	0.186

Table 13: AIC Comparison of Nested Models

Model	AIC
Model: log(income)	84.56
Model: log(income) + literacy_rate	84.63

Conclusion and Summary

The goal of this analysis was to investigate the relationship between income, education levels, and crime rates across different regions/countries. The analysis used World Bank API data to retrieve GDP per capita, literacy rate, and intentional homicide rates. Some key findings were: Income and Literacy Rate: A strong positive correlation of 0.86 was found between income and literacy rate, confirming that wealthier countries tend to have higher literacy levels. Income and Crime Rate: A weak negative correlation of -0.24 suggested that higher-income countries generally have lower crime rates, but the relationship is not strong. Literacy and Crime Rate: A moderate negative correlation -0.40 showed that higher literacy rates are associated with lower crime rates, but other factors may also influence crime levels. Distribution of Crime Rates: Low-income countries showed higher crime rates with less variation. Middle-income countries had the highest variation in crime which meant that income alone does not fully explain crime trends. High-income countries exhibited generally lower crime rates but still showed some variation. Boxplots & Density Plots: These visualizations confirmed that crime rates tend to be higher in lower-income and lower-literacy countries, but exceptions exist. To validate assumptions and better model the relationships, the Shapiro-Wilk test was used to assess the normality of variables. Log-transformations were applied to income and crime rate due to their strong right-skew. Multiple linear regression revealed that literacy rate was a statistically significant predictor of crime rate, whereas log-transformed income was not. A weighted regression (to correct for heteroscedasticity) confirmed the significance of literacy, reinforcing its potential role in reducing crime. Model comparison using ANOVA and AIC further evaluated whether literacy meaningfully improved model performance beyond income alone. The results indicated that although literacy was significant in the regression model, its addition did not significantly improve overall model fit which meant an overlapping explanatory power with income or limitations due to multicollinearity. These results suggest that while both income and education are related to crime, education reflected by literacy rate may have a more direct and statistically strong association with crime reduction. Still, the presence of moderate multicollinearity and variation across income groups implies that no single factor can fully explain international crime patterns. Further work might integrate other social and governance variables such as unemployment, urbanization, inequality to improve explanatory power and guide evidence-based policy recommendations.

Limitations

This analysis has several limitations that should be acknowledged. First, the dataset used was relatively small, consisting only of countries/regions with complete data on income, literacy rate, and crime rate for the year 2015. As a result, more advanced modeling techniques such as machine learning were not appropriate.

A small sample size limits both the generalizability of the findings and the statistical power to detect subtle relationships or nonlinear patterns. Second, the analysis was cross-sectional, capturing data from a single year. This prevents any assessment of causality or changes over time, which are often critical when studying social and economic dynamics. Third, some potential confounding variables were not included such as unemployment rate, level of urbanization, quality of governance, or police expenditure which could influence crime rates. Additionally, the moderate multicollinearity observed between income and literacy rate may have masked the individual effects of these predictors in the regression models. Finally, due to limitations in available geospatial data, regional visualizations were constrained and could not be fully explored.